

Quiz (Habanero)**Q1.** Evaluate the expression $3 \cdot 2 + 11 - 4$.

- (A) 14
- (B) 15
- (C) 16
- (D) 17
- (E) 18

Q2. What is the value of x in the expression $x + 5 = 11$?

- (A) 5
- (B) 6
- (C) 7
- (D) 8
- (E) 9

Q3. The temperature in a city is given by $T = 2h - 5$, where h is the number of hours since midnight. Find T when $h = 4$.

- (A) 3
- (B) -1
- (C) 3
- (D) 5
- (E) -3

Q4. The rainfall in a region is given by $R = 0.5x + 2$, where x is the number of days. Find R when $x = 6$.

- (A) 4
- (B) 5
- (C) 6
- (D) 7
- (E) 8

Q5. Simplify the expression $2(x + 3) + x - 1$.

- (A) $2x + 5$
- (B) $3x + 6$
- (C) $3x + 5$
- (D) $2x + 6$
- (E) $x + 5$

Q6. The cost of removing pollutants from a lake is given by $C = 200 + 50x$, where x is the number of days. Find C when $x = 3$.

- (A) 350
- (B) 400
- (C) 500
- (D) 550
- (E) 600

Q7. The population of a species is given by $P = 100 + 20t - 5t^2$, where t is the number of years. Find P when $t = 2$.

- (A) 100
- (B) 120
- (C) 140
- (D) 160
- (E) 180

Q8. Evaluate the expression $(2 + 3) \cdot (4 - 1)$.

- (A) 10
- (B) 12
- (C) 14
- (D) 16
- (E) 18

Open Ended Questions (FRQ)**Q9.** A water tank can hold 1000 liters of water. The water level in the tank is given by $L = 2t + 50$, where t is the number of hours since the tank was filled. Find the time t when the tank is half full.**Q10.** The temperature in a city is given by $T = 5x - 2$, where x is the number of hours since midnight. Find the expression for the temperature T when it is 3 hours ahead.**Chef's Tips**

- Always check the units of the answer.
- Read the question carefully before answering.
- Use the correct order of operations.